

MTH203: Multivariate Calculus

Monsoon 2025

Name: _____

Roll No.: _____

Tutorial Quiz 6

Question:

1. Find the length of the curve

$$\mathbf{r}(t) = (\cos t + t \sin t)\mathbf{i} + (\sin t - t \cos t)\mathbf{j}, \quad \sqrt{2} \leq t \leq 2 \quad (1)$$

2. For the helix

$$\mathbf{r}(t) = (3 \sin t)\mathbf{i} + (3 \cos t)\mathbf{j} + 4t\mathbf{k}$$

find the unit tangent vector $\mathbf{T}$ and the curvature κ . (2)

Please note: There is no fractional marking. Only integer marking.